Assignment 3

OPTI 544 Quantum Optics

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Exercise 1

a) Assuming a green laser that combine to generate a pair of entangled red and blue photons. The E-field and operators are:

$$\boldsymbol{E}_z = \hat{z}\left[\mathcal{E}_G e^{i(kx-\omega_G t)} + \mathcal{E}_G^* e^{-i(kx-\omega_G t)}\right],$$
$$\hat{\boldsymbol{E}}_z = i\sum_{\boldsymbol{k}_R}\sqrt{\frac{\hbar\omega_R}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k}_R,z}e^{i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)} - \hat{a}^\dagger_{\boldsymbol{k}_R,z}e^{-i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)}\right],$$
$$\hat{\boldsymbol{E}}_z = i\sum_{\boldsymbol{k}_B}\sqrt{\frac{\hbar\omega_B}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k}_B,z}e^{i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)} - \hat{a}^\dagger_{\boldsymbol{k}_B,z}e^{-i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)}\right].$$

Using these definitions in the product integrand of the Hamiltonian yields

$$\boldsymbol{E}_z\boldsymbol{E}_z\hat{\boldsymbol{E}}_z\hat{\boldsymbol{E}}_z = -\frac{\hbar}{2\varepsilon_0 V}\hat{z}\cdot\hat{z}\sum_{\boldsymbol{k}_B,\boldsymbol{k}_R}\sqrt{\omega_R\omega_B}\left[\mathcal{E}_G e^{i(kx-\omega_G t)} + \mathcal{E}_G^* e^{-i(kx-\omega_G t)}\right]\left[\mathcal{E}_G e^{i(kx-\omega_G t)} + \mathcal{E}_G^* e^{-i(kx-\omega_G t)}\right]$$
$$\left[\hat{a}_{\boldsymbol{k}_R,z}e^{i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)} - \hat{a}^\dagger_{\boldsymbol{k}_R,z}e^{-i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)}\right]\left[\hat{a}_{\boldsymbol{k}_B,z}e^{i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)} - \hat{a}^\dagger_{\boldsymbol{k}_B,z}e^{-i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)}\right].$$

By distributing all the terms above, we will have different combinations of frequency and wavevectors. In particular,

$$\omega:\ e^{i(\pm 2\omega_G \pm \omega_R \pm \omega_B)t} \quad \wedge \quad \boldsymbol{k}:\ e^{i[\pm 2kx + (\pm\boldsymbol{k}_R \pm \boldsymbol{k}_B)\cdot\boldsymbol{r}]}.$$

The only terms that are slow oscillating are the ones that are nearly zero, obtaining the energy and momentum conservation:

$$\pm 2\omega_G \mp (\omega_R + \omega_B) = 0 \Longrightarrow \text{Energy conservation}: 2\omega_G = \omega_R + \omega_B,$$
$$\pm 2kx \mp (\boldsymbol{k}_R + \boldsymbol{k}_B) \Longrightarrow \text{Momentum conservation}: 2kx = \boldsymbol{k}_R + \boldsymbol{k}_B.$$

The difference of blue and red frequencies is still in the optical frequency. The Hamiltonian then reduces to

$$\hat{H}^{(3)}_{NL,1} = \frac{\hbar\sqrt{\omega_R\omega_B}}{2}\chi^{(3)}_{zzzz}\left[\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z}\right].$$

b) The evolution operator applied to the vaccum state of R and B light $|0\rangle_R |0\rangle_B = |\{0\}\rangle$ is:

$$\hat{U}(t,0) = e^{-i\hat{H}^{(3)}_{NL,1}t/\hbar} |\{0\}\rangle = e^{-i\left\{\frac{\hbar\sqrt{\omega_R\omega_B}}{2}\chi^{(3)}_{zzzz}\left[\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z}+\mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z}\right]\right\}t/\hbar} = |\psi_{out}\rangle$$

Expanding the exponential to the first-order Taylor series yields:

$$\left\{\mathbb{1} - i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzzz}}{2}\left[\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z}\right]t\right\}|\{0\}\rangle .$$

By defining the squeezing parameter ξ_z as:

$$\xi_z = -i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzzz}}{2}\mathcal{E}_G^2 t,$$

we have:

$$\left[\mathbb{1} + \xi_z\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \xi_z^*\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z}\right]|\{0\}\rangle = |\{0\}\rangle + \xi_z|1\rangle_{R_z}|1\rangle_{B_z} .$$

c) The process for the new term is esentially the same, with the change of the last two indices to y polarization. The Hamiltonian for this term is:

$$\hat{H}^{(3)}_{NL,2} = \frac{\hbar\sqrt{\omega_R\omega_B}}{2}\chi^{(3)}_{zzyy}\left[\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y}\right]$$

The total Hamiltonian is the sum of the individual Hamiltonian:

$$\hat{H}^{(3)}_{NL} = \hat{H}^{(3)}_{NL,1} + \hat{H}^{(3)}_{NL,2}$$
$$\hat{H}^{(3)}_{NL} = \frac{\hbar\sqrt{\omega_R\omega_B}}{2}\left[\chi^{(3)}_{zzzz}\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z} + \chi^{(3)}_{zzyy}\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y}\right].$$

The first-order Taylor expansion of the evolution operator is:

$$U(t,0) = e^{-i\hat{N}^{(3)}_{NL}t/\hbar} \approx \mathbb{1} + \xi_z\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \xi_z^*\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z} + \xi_y\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \xi_y^*\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y},$$

where the squeezing parameters are:

$$\xi_z = -i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzzz}}{2}\mathcal{E}_G^2 t, \quad \text{and} \quad \xi_y = -i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzyy}}{2}\mathcal{E}_G^2 t.$$

Application of the evolution operator in the vaccum state produces:

$$\left[\mathbb{1} + \xi_z\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \xi_z^*\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z} + \xi_y\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \xi_y^*\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y}\right]|\{0\}\rangle = |\{0\}\rangle + \xi_z|1\rangle_{R_z}|1\rangle_{B_z} + \xi_y|1\rangle_{R_y}|1\rangle_{B_y} .$$

Generating in that way, two polarization entangled photons.

Exercise 2

a) First, the eigenequation of the number operator $\hat{n}$ and the definition of the thermal density matrix are:

$$\hat{n}|n\rangle = n|n\rangle , \quad \text{and} \quad \hat{\rho}_{th} = \sum_n P_n |n\rangle\langle n| .$$

The expectation value of the number of photons is:

$$\langle \hat{n} \rangle = \mathrm{Tr}\left[\hat{n}\hat{\rho}_{th}\right] = \sum_n \langle n|\hat{n}\hat{\rho}_{th}|n\rangle = \sum_{n,n'} \langle n|\hat{n}P_{n'}|n'\rangle\langle n'|n\rangle = \sum_{n,n'} n'P_{n'}|\langle n|n'\rangle|^2 = \sum_n nP_n.$$

Now, we put the Boltzman distribution in the last summation:

$$\langle \hat{n} \rangle = \sum_n n\left[\frac{e^{-(n+1/2)\hbar\omega/(k_BT)}}{\sum_{m=0}^{\infty} e^{-(m+1/2)\hbar\omega/(k_BT)}}\right] = \sum_n n\frac{e^{-n[\hbar\omega/(k_BT)]}}{\sum_{m=0}^{\infty} e^{-m[\hbar\omega/(k_BT)]}} = \frac{1}{\sum_{m=0}^{\infty} e^{-mx}}\sum_n ne^{-nx},\ x = \frac{\hbar\omega}{k_BT}.$$

We note that the summation in n can be expressed in terms of a geometric function in the following way

$$\sum_{n=0}^{\infty} ne^{-nx} = -\frac{d}{dx}\left[\sum_{n=0}^{\infty} e^{-nx}\right] = -\frac{d}{dx}\left[\frac{1}{1-e^{-x}}\right] = \frac{e^{-x}}{(1-e^{-x})^2}.$$

Then,

$$\langle \hat{n} \rangle = \frac{1}{\sum_{m=0}^{\infty} e^{-mx}}\frac{e^{-x}}{(1-e^{-x})^2} = (1-e^{-x})\frac{e^{-x}}{(1-e^{-x})^2} = \frac{e^{-x}}{1-e^{-x}} = \frac{1}{e^x-1} = \frac{1}{e^{\frac{\hbar\omega}{k_BT}}-1}.$$

which is the Planck radiation distribution.

b) By evaluating the above result, and knowing that

$$h \approx 6.6261 \cdot 10^{-34}\ m^2kg/s, \quad \text{and} \quad k_B \approx 1.3806 \cdot 10^{-23}\ m^2kgs^{-2}K^{-1}.$$

The average number of photon as a function of the absolute temperature is:

$$\langle \hat{n} \rangle(T) = \frac{1}{e^{\frac{[6.6261 \cdot 10^{-34}/(2\pi)][(2\pi)(3 \cdot 10^8)/(500 \cdot 10^{-9})]}{(1.3806 \cdot 10^{-23})K}} - 1} = \frac{1}{e^{28797/K}-1}.$$

At $300K$ and $6000K$ we have

$$\langle \hat{n} \rangle(300) = \frac{1}{e^{28797/300}-1} = 2.1 \cdot 10^{-42}, \quad \text{and} \quad \langle \hat{n} \rangle(6000) = \frac{1}{e^{28797/6000}-1} = 8.3 \cdot 10^{-3}.$$